## Investigation: INEPAR EQUIPAMENTOS E MONTAGENS S/A (CVM code 17868), fiscal year 2012, as it was on 2014-04-30

Statistical signal based on public financial statements. Not an accusation, not investment advice.

## Company

Sector: Máquinas, Equipamentos, Veículos e Peças (origin: FCA); group: non-financial.
nenhum código de negociação no FCA entregue até a data (nos dados abertos, o campo só vem preenchido a partir do FCA de 2018)

## Document used

2012 DFP, version 1, filed on 2013-03-28; origin: open data; type: individual; original scale: MIL.
Versions known on the date: 1 (2013-03-28).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=25668

## Possible manipulation (engine v1, study 1)

Beneish M-score
- Value: −2.326 (index)
- Alert by the study 1 rule: no
- Rule: M > −1.78 (pre-registration, section 4)
- Percentile in sector and year: 56 (n = 17)
- Percentile in group and year: 67 (n = 358)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

DSRI, days' sales in receivables index
- Value: 1.088 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 64 (n = 21)
- Percentile in group and year: 66 (n = 449)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

GMI, gross margin index
- Value: 0.844 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 12 (n = 21)
- Percentile in group and year: 13 (n = 463)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

AQI, asset quality index
- Value: 1.016 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 60 (n = 21)
- Percentile in group and year: 55 (n = 460)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

SGI, sales growth index
- Value: 0.894 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 12 (n = 21)

Percentile in group and year: 12 (n = 464)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DEPI, depreciation index
Value: 1.132 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 62 (n = 17)
Percentile in group and year: 87 (n = 377)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

SGAI, SG&A expenses index
Value: 0.594 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 7 (n = 21)
Percentile in group and year: 5 (n = 457)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

TATA, total accruals to total assets
Value: 0.031 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 80 (n = 22)
Percentile in group and year: 83 (n = 479)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

LVGI, leverage index
Value: 0.960 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 31 (n = 21)
Percentile in group and year: 31 (n = 472)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DPO jump
Value: 6.7 days
Alert by the study 1 rule: no
Rule: DPO change in the year's top decile among non-financial companies (≥ 90th percentile of computable values on the date: 32.0 days)
Percentile in sector and year: 62 (n = 20)
Percentile in group and year: 70 (n = 432)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium
Alternative explanations: Bargaining power with suppliers and a change in purchase mix (decision of 2026-11-07)

Implied interest rate minus average CDI
Value: 6.53 p.p.
Alert by the study 1 rule: no
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: 39 (n = 22)
Percentile in group and year: 47 (n = 431)
The company's own history: fewer than 5 previous fiscal years with a value

Confidence in the data: medium
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets
Value: 0.032 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: 80 (n = 22)
Percentile in group and year: 83 (n = 478)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

## Financial risk (context only)

Altman Z'' (financial risk, context only)
Value: 1.600 (index)
Components: X1 = −0.049; X2 = 0.233; X3 = −0.005; X4 = 1.137 (indices)
Percentile in sector and year: 41 (n = 23)
The company's own history: fewer than 5 previous fiscal years with a value
- No cut-off zone: the available source does not define zones for Z''
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

## Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 3/9 (8 computable)
Signals:
- Return on assets (ROA): 1
- Operating cash flow on assets (CFO): 0
- Change in ROA (ΔROA): 0
- Cash above earnings (ACCRUAL): 0
- Change in leverage (ΔLEVER): 1
- Change in current ratio (ΔLIQUID): 0
- No equity issuance (EQ_OFFER): —
- Change in gross margin (ΔMARGIN): 1
- Change in asset turnover (ΔTURN): 0
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement

## Changes

Highlighted in the following year's comparative: 4 (threshold of R$7.2 million, 1% of total assets of the version used). Values in R$ million:
BPA 1 Ativo Total: 718.1 in the version used (v1, filed on 2013-03-28); 437.2 in the next DFP's comparative (v1, filed on 2014-03-28); difference of −280.9. Different value in the following year's comparative (restatement or reclassification).
BPP 2.03 Patrimônio Líquido: 382.0 in the version used (v1, filed on 2013-03-28); 167.4 in the next DFP's comparative (v1, filed on 2014-03-28); difference of −214.6. Different value in the following year's comparative (restatement or reclassification).
BPP 2.02 Passivo Não Circulante: 280.6 in the version used (v1, filed on 2013-03-28); 214.3 in the next DFP's comparative (v1, filed on 2014-03-28); difference of −66.2. Different value in the following year's comparative (restatement or reclassification).
DRE 3.11 Lucro/Prejuízo do Período: 20.0 in the version used (v1, filed on 2013-03-28); −5.0

in the next DFP's comparative (v1, filed on 2014-03-28); difference of −25.0. Different value in the following year's comparative (restatement or reclassification).

Highlighted: differences in totals and key lines (total assets, current liabilities, non-current liabilities, equity, net revenue, net income and operating cash flow) above 1% of total assets of the version used. The other differences (reclassifications between lines and smaller differences) stay in the full list.

Full list: 30 accounts above the R$1,000 tolerance (group a, totals and key lines: 4; group b, reclassifications: 26), in the annex at the end.

## What was known on the date

- DFP 2010, v1: 2011-03-31
- FCA 2012, v1: 2012-03-26
- DFP 2011, v1: 2012-04-02
- FCA 2012, v2: 2012-05-15
- DFP 2012, v1: 2013-03-28
- DFP 2013, v1: 2014-03-28

DFP and FCA filed with the CVM by the reference date; ITR and material facts come in later phases.

## Data quality

53 applicable checks; 0 failed.

## Auditor's report and auditor (factual section; no alert)

Auditor's report: Com Ressalva (qualified).

no opinion excerpt was found in the text

Auditor in the 2011 FCA: Martinelli Auditores.

Auditor in the 2012 FCA: Baker Tilly Brasil Auditores Independentes S/S., Martinelli Auditores.

Auditor in the 2013 FCA: Baker Tilly Brasil Auditores Independentes S/S..

Auditor changes (one line per change, canonical name by the firm's CVM code):

- 2012: Martinelli Auditores -> Baker Tilly Brasil Auditores Independentes S/S. (transition: two firms in the 2012 FCA). The outgoing firm audited 8 consecutive fiscal years (2004 to 2011), by the start and end dates in the FCA.

Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

## How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

## Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

## Limitations

- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 38 DFP lines, each with document, version, filing date, account and location.

## Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a E). Generated at 2026-11-10T19:32:30+00:00.

## Today's information (outside the reference date)

Registry status: ATIVO. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.

## Annex: full list of differences in the following year's comparative (R$ million)

(a) BPA 1 Ativo Total: 718.1 in the version used, 437.2 in the comparative (difference of −280.9)
(a) BPP 2.03 Patrimônio Líquido: 382.0 in the version used, 167.4 in the comparative (difference of −214.6)
(a) BPP 2.02 Passivo Não Circulante: 280.6 in the version used, 214.3 in the comparative (difference of −66.2)
(a) DRE 3.11 Lucro/Prejuízo do Período: 20.0 in the version used, −5.0 in the comparative (difference of −25.0)
(b) BPA 1.02 Ativo Não Circulante: 697.6 in the version used, 416.7 in the comparative (difference of −280.9)
(b) BPP 2 Passivo Total: 718.1 in the version used, 437.2 in the comparative (difference of −280.9)
(b) BPA 1.02.01 Ativo Realizável a Longo Prazo: 566.4 in the version used, 326.2 in the comparative (difference of −240.1)
(b) BPA 1.02.01.01 Aplicações Financeiras Avaliadas a Valor Justo: 247.2 in the version used, 7.0 in the comparative (difference of −240.1)
(b) BPA 1.02.01.01.03 Títulos e Valores Mobiliários: 247.2 in the version used, 7.0 in the comparative (difference of −240.1)
(b) BPP 2.03.04 Reservas de Lucros: 167.0 in the version used, −47.7 in the comparative (difference of −214.6)
(b) BPP 2.03.04.01 Reserva Legal: 167.0 in the version used, −47.7 in the comparative (difference of −214.6)
(b) BPP 2.02.02 Outras Obrigações: 121.9 in the version used, 55.7 in the comparative (difference of −66.2)
(b) BPP 2.02.02.02 Outros: 86.8 in the version used, 20.6 in the comparative (difference of −66.2)
(b) BPP 2.02.02.02.03 Impostos e Contribuições a Recolher: 86.8 in the version used, 20.6 in the comparative (difference of −66.2)
(b) BPA 1.02.02 Investimentos: 84.3 in the version used, 43.5 in the comparative (difference of −40.7)
(b) BPA 1.02.02.01 Participações Societárias: 84.3 in the version used, 43.5 in the comparative (difference of −40.7)
(b) BPA 1.02.02.01.01 Participações em Coligadas: 84.3 in the version used, 43.5 in the comparative (difference of −40.7)
(b) DFC (indirect method) 6.01.01.07 Lucro Liquido do Exercício: 20.0 in the version used, −5.0 in the comparative (difference of −25.0)
(b) DRE 3.07 Resultado Antes dos Tributos sobre o Lucro: 19.6 in the version used, −5.4 in the comparative (difference of −25.0)
(b) DRE 3.09 Resultado Líquido das Operações Continuadas: 20.0 in the version used, −5.0 in the

comparative (difference of −25.0)
(b) DRE 3.06.01 Receitas Financeiras: 47.4 in the version used, 22.6 in the comparative (difference of −24.8)
(b) DFC (indirect method) 6.01.01 Caixa Gerado nas Operações: −7.4 in the version used, −27.8 in the comparative (difference of −20.4)
(b) DFC (indirect method) 6.01.02 Variações nos Ativos e Passivos: 5.0 in the version used, 25.4 in the comparative (difference of 20.4)
(b) DRE 3.06 Resultado Financeiro: 23.1 in the version used, 2.6 in the comparative (difference of −20.4)
(b) DFC (indirect method) 6.01.01.03 Equivalência Patrimonial: −0.6 in the version used, 3.9 in the comparative (difference of 4.6)
(b) DRE 3.04 Despesas/Receitas Operacionais: −4.2 in the version used, −8.8 in the comparative (difference of −4.6)
(b) DRE 3.04.06 Resultado de Equivalência Patrimonial: 0.6 in the version used, −3.9 in the comparative (difference of −4.6)
(b) DRE 3.05 Resultado Antes do Resultado Financeiro e dos Tributos: −3.5 in the version used, −8.0 in the comparative (difference of −4.6)
(b) DFC (indirect method) 6.01.02.08 Impostos e Contribuições a Recolher: 5.6 in the version used, 1.3 in the comparative (difference of −4.3)
(b) DRE 3.06.02 Despesas Financeiras: −24.3 in the version used, −19.9 in the comparative (difference of 4.3)